

Alexandre LOU-POUEYOU

High Performance Computing Internship

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Profile

Computer science student with a strong interest in high-performance computing and data-intensive systems. I am particularly interested in how data movement, storage hierarchies, and access patterns impact the performance and scalability of large-scale applications.

My background includes parallel computing (MPI, OpenMP, CUDA), systems programming, and performance analysis, with a focus on understanding and optimizing complex interactions between software and hardware in distributed environments.

Education

- 2024–Present **Master's Degree in Computer Science — HPC & Data Science**, *University of Bordeaux / Enseirb*, Talence
Relevant coursework: Parallel Architecture Programming (PAP), Network & Protocols, Distributed Systems, System Administration
- 2021–2024 **Bachelor's Degree in Computer Science**, *University of Bordeaux*, Talence
Strong foundation in algorithms, data structures, and low-level programming (C)

Experience

- Feb 2026 – **HPC Internship**, *Inria Center of the University of Bordeaux*, Talence
July 2026
 - Performance analysis and optimization of I/O in the Gysela plasma simulation code
 - Investigation of I/O bottlenecks in data-intensive HPC workloads using distributed storage systems
 - Design and evaluation of parallel I/O strategies using BeeGFS
 - Analysis of scalability and checkpointing overheads (hundreds of GB scale)
 - Study of compute–I/O overlap in large-scale scientific simulations
 - Benchmarking and performance profiling on HPC infrastructure
- 2026–2027 **Network & System Administration**, *Association Label[i]*, Talence
 - Administration of VPS infrastructure (deployment, monitoring, maintenance)
 - Configuration and management of network services (web, reverse proxy, monitoring tools)
 - Implementation of security practices and service isolation
- 2023–2025 **Seasonal Job — Payroll Management**, *Sodexo*, Bordeaux
Data processing, document management, and rigorous handling of administrative workflows
- 2022–2025 **President / Secretary**, *Association Label[i]*, Talence
Project management, team coordination, and organization of computer science events

Academic & Personal Projects

- Chess Engine (C++) Development of an optimized chess engine with a modular architecture. Focus on performance optimization, game tree search, and parallelization strategies (CPU/GPU).
- Portfolio Website (Vite/Vue, SCSS) Personal portfolio to showcase projects and technical progress. Built with modern web technologies
- Takuzu-2077 (C) Implementation of the Takuzu puzzle game with both console and graphical interfaces.

Skills

HPC & Parallel MPI · OpenMP · CUDA · OpenCL · Vectorization · I/O Optimization · Slurm

Programming C · C++ · Python · Cython · Java · TypeScript · Shell (Bash)

Systems & Tools Linux · Docker · Git · CI/CD · CMake · GDB · Valgrind · Flask

Core Interests Distributed Systems · Performance Profiling · Data-Intensive Applications

Languages

French Native

English Fluent (C1)

Japanese Beginner (A2)

Hobbies

Running (marathon), Strength training, Guitar, Reading, Chess